E.69 standalone proof review

Statement E.69 (Proved). The Gaussian central layer and its exact common finite fibres. *Let H_3 be the exponent-three Heisenberg group, written in logarithmic coordinates as*

$$(v,c)(w,e) = (v+w, c+e+\tfrac{1}{2}\det(v,w)), \qquad v,w \in \mathbf{F}_3^2, \quad c,e \in \mathbf{F}_3. \tag{E.69.1}$$

Here $1/2 = 2$. Let d have order four and act by $d(v,c)d^{-1} = (Sv,c)$, with the matrix S of E.67.2. The actual Gaussian triangle quotient

$$\widetilde{\mathrm{G}} = H_3 \rtimes \langle d \rangle, \qquad x = (e_1,0)d, \quad y = (-e_2,0)d^2, \quad z = d \tag{E.69.2}$$

has order 108. Keep the Mordell quotient M of E.67 unchanged. Evaluate each word in the same six ordered pairs E.67.5 in each of these two groups, and denote this actual twelve-evaluation homomorphism by $\widetilde{\Theta}$.

Define two automorphisms of H_3 by

$$A(v,c) = (-v, c+v_1+v_2), \qquad B(v,c) = (-Sv, c+v_1+v_2). \tag{E.69.3}$$

For $N = N_B$ and $K_6 = \overline{N'N^6}$, the complete correction image is

$$\begin{aligned} L := \widetilde{\Theta}(N') = \widetilde{\Theta}(K_6) = \mathcal{M} \times \widetilde{\mathcal{G}}, \\ \widetilde{\mathcal{G}} = \{(u,v,t,A(t),B(v),B(u)) : u,v,t \in H_3\} \simeq H_3^3. \end{aligned} \tag{E.69.4}$$

In particular $|L| = 3^{12}$. The actual projection which forgets the six Gaussian central coordinates induces

$$L^{\mathrm{ab}} \xrightarrow{\sim} \mathcal{M} \times \mathcal{G}, \qquad L' \simeq \mathbf{F}_3^3. \tag{E.69.5}$$

The right side and its source map are precisely those of E.67 and E.68.

There are four automorphisms of $\widetilde{\mathrm{G}}$ which fix x, carry y,z to conjugates by elements of H_3, and preserve the positive primitive boundary commutator. They are conjugation by $1, x, x^2, x^3$. They have 36 pairs of such frames. Exactly twelve framed six-tuples belong to $\widetilde{\mathcal{G}}$: if $h_y, h_z \in H_3$ are the frames, their equation is

$$(h_z)_c = (h_z)_1 + (h_z)_2, \qquad (h_y)_c \text{ arbitrary.} \tag{E.69.6}$$

Thus each of the four E.67 Gaussian actions has exactly three framed lifts in the actual correction image.

Retain the E.66 map $\Psi : N' \to Z$ and exactly the 24 E.17 origins used in E.68. The complete common image is

$$(\Psi, \widetilde{\Theta})(N') = Z \times L, \qquad |Z \times L| = 189\,321\,481\,108\,517\,289\,984. \tag{E.69.7}$$

Every one of E.68's 1152 finite combinations has exactly three lifts to these specified Gaussian framed targets. All 3456 combinations have saved actual N'-word preimages. For one such correction u over an origin f_i, the complete word fibre inside f_iN' is

$$f_i u\big(N' \cap \ker \Psi \cap \ker \widetilde{\Theta}\big). \tag{E.69.8}$$

These conclusions concern the displayed finite maps and their full correction-image fibres. They retain the Mordell quotient of E.67 and do not assign a full-GT lift or generation in an additional unevaluated source.

Proof. We identify the quotient, its complete source image and the common word fibres in this order.

The actual Gaussian quotient and its frames. In the original quartic triangle, put $X = x^2y$ and $Y = z^2y$. After killing the central fourth power of z, the exact free-by-cyclic presentation of E.64 gives

$$dXd^{-1} = Y^{-1}, \quad dYd^{-1} = X, \quad x = Xd, \quad y = Y^{-1}d^2, \quad z = d. \tag{E.69.9}$$

Pass the free kernel $\langle X, Y\rangle$ to its class-two exponent-three quotient. Its collection map is exactly (E.69.1), with $X = (e_1, 0)$ and $Y = (e_2, 0)$. Thus (E.69.2) is a quotient of the actual triangle, with no change of peripheral words. Direct collection gives

$$xyz = 1, \qquad y^2 = z^4 = 1, \qquad x^4 = [X, Y^{-1}] = [X, Y]^{-1}. \tag{E.69.10}$$

Forgetting the central coordinate gives the literal Gaussian affine quotient of E.67.3.

The finite model lists all 108 elements and verifies the multiplication law, all inverses and all 108^3 associativity triples. For each of the 27^2 pairs (h_y, h_z) it forms

$$y_* = h_y y h_y^{-1}, \qquad z_* = h_z z h_z^{-1} \tag{E.69.11}$$

and tests $xy_*z_* = 1$. It then propagates $x \mapsto x$, $y \mapsto y_*$ along every edge of the full 108-element Cayley graph. Agreement on every edge proves a homomorphism; 108 distinct images prove it is an automorphism. It also checks the image of z and the positive commutator. Exactly four distinct automorphisms survive, each with nine frame pairs; direct comparison with all conjugations by x^j, $0 \le j < 4$, identifies them as stated. Their actual six word values are

$$(h_y^{-1}, h_y, h_z, h_z^{-1}, h_z^{-1}h_y, h_y^{-1}h_z). \tag{E.69.12}$$

This ordering retains both three-cycle and inverse equations, including the central terms.

The complete correction image. Use the unchanged 97 literal free generators of N_B. Every one of their twelve values has even deck exponent. The even-deck subgroup of M has exponent three, and the even-deck subgroup of $\widetilde{\mathrm{G}}$ has exponent six. These exponent assertions follow from the group laws for every element of those subgroups. Hence all N^6 values vanish and the two images in (E.69.4) agree.

On the deck-zero part use the twelve Mordell translation coordinates followed by six Gaussian triples (v_1, v_2, c), a total of thirty coordinates. The program evaluates all $\binom{97}{2} = 4656$ commutators, closes their span under conjugation by all 97 actual generators and under Lie brackets, and retains literal word preimages for every row. The initial commutator rank is twelve; all subsequent normal-conjugation and bracket checks remain in that same span. The following formula realizes each linear row addition by an actual word in this class-two exponent-three group:

$$\log(ab[a, b]) = \log(a) + \log(b). \tag{E.69.13}$$

Indeed the Baker–Campbell–Hausdorff coefficient is $1/2 = 2$, and $2 + 1 = 0$ in $\mathbf{F}_3$; multiplication by two is inversion. Thus no row operation replaces the source word map by an abstract linear surrogate. The closed normal generation of N' by the pairwise commutators proves that this is its complete image.

Both A and B in (E.69.3) preserve the law (E.69.1): their linear parts have determinant one and their central corrections are linear. The twelve-dimensional span just computed is exactly the direct product in (E.69.4). This equality is checked in both directions against its twelve displayed coordinate basis vectors. The three brackets of the two generators in each H_3 factor form the entire three-dimensional derived subgroup. Their common kernel under the E.67 projection also has dimension three. This proves the precise abelianization isomorphism (E.69.5), including its source projection.

The same nine-dimensional Gaussian image is already the image of $\widehat{F}'$. To check this assertion with its exact map, start with the single literal $[x, y]$, close its Gaussian span under conjugation by the two free-source generators and under brackets, and compare it in both directions with the nine Gaussian basis vectors in (E.69.4). The resulting rank is nine. Consequently each retained origin also has its Gaussian value in the displayed graph. Substitution of the 36 frame pairs (E.69.12) in that graph yields exactly (E.69.6), with twelve solutions. For every actual E.17 word, the correction is evaluated as $f_i^{-1}t$ in the group law, including its central cross term. The complete source membership test gives the same twelve solutions. Their Mordell frame is the uniquely specified one already proved in E.67.

The common quotient with the retained distribution source. Let J be the actual image of $(\Psi, \widetilde{\Theta}) : N' \to Z \times L$. Both projections are onto. Goursat's construction gives a common quotient of Z and the finite three-group L. A nontrivial finite three-group has a nontrivial cyclic quotient of order three. Its composition with L factors through L^{ab}. But (E.69.5) identifies this particular abelianization map with E.67, and E.68 proves that its actual common image with Ψ is the full product. Such a cyclic common quotient cannot occur. Thus the common quotient is trivial and (E.69.7) follows.

For explicit common preimages, the new certificate reads E.68's complete Schreier presentation without changing it. It obtains six words in N' whose Ψ values are one, whose Mordell values are zero, and whose Gaussian E.67 values are the six successive horizontal basis vectors of H_3^3. Their three paired commutators have zero old E.67 values and are exactly the three new central basis vectors. Their old-source identities are replayed on the entire 5331-point source. In fact all three commutators are the identity on all 5223 old induced points, which is stronger than the retained Ψ assertion.

Start with each of the 1152 saved E.68 words and evaluate it in the actual new twelve groups. For each of the three central lifts of its Gaussian frame, multiply by the necessary product of the three just-constructed central words. All 3456 resulting words are replayed in the new groups; their old values are retained by the checked central identities. The finite target counts and full word cosets (E.69.8) follow.

The four programs share the prefix `bc_gaussian_heisenberg_`: the suffixes `model`, `source`, `structure` and `joint` record respectively the actual finite quotient, complete normal source, exact graph and abelianization, and all common word preimages. Their saved arrays include the literal word DAG, its source generators, all central preimages and every successful finite target. □